

Esta é uma cópia de uma conversa entre ChatGPT e Anônimo.
Denunciar conteúdo

Arquivo carregado

Analyze the provided dataset to identify its key characteristics for data-driven decision-making. Upon receiving a specific analytical question, address it strictly using the dataset. Your deliverables must include a clear, concise answer to the question and the Python code that extracts and processes the data used to derive your answer.

The dataset contains **167,826 records** and **9 columns**, focused on HIV epidemiology in children and adolescents. Here's a summary of its key characteristics:

Key Columns:

- **Country/Region:** Geographic area.
- **UNICEF_Region:** UNICEF-designated regional grouping (some missing values).
- **Indicator:** Type of HIV-related measure (e.g., incidence rate).
- **Year:** Year of measurement.
- **Sex:** Gender (Both, Male, Female).
- **Age:** Age group (e.g., 0-14, 15-19).
- **Value, Lower, Upper:** Estimated value and confidence intervals (often stored as strings, e.g., "<0.01").

This structure is well-suited for time-series, regional, and demographic analysis of HIV trends.

You're now ready to ask any specific analytical question you'd like answered using this dataset.

Using historical data (2005–2023), calculate prediction intervals for HIV case growth in 2024, incorporating median estimates and uncertainty bounds.

To estimate **HIV case growth**, the most relevant indicator is:

"Estimated number of annual new HIV infections"

This represents new cases per year and is ideal for trend analysis and forecasting.

I'll now filter the dataset to this indicator and compute the **median annual values** from 2005–2023, then use that to forecast **prediction intervals for 2024** based on historical uncertainty bounds.

Forecast for 2024: Estimated New HIV Infections (Global, Ages 0–19)

Based on historical trends (2005–2023), the **predicted number of new HIV infections in 2024** is:

- **Median estimate:** ~360,754
- **Lower bound** (95% prediction interval): ~342,791
- **Upper bound** (95% prediction interval): ~368,958

This forecast reflects a linear projection using historical median values and associated uncertainty.

Python Code Used:

```
python
from sklearn.linear_model import LinearRegression
import numpy as np

# Prepare data for modeling
X = df_yearly_summary[['Year']]
y_median = df_yearly_summary['median_value']
y_lower = df_yearly_summary['lower_bound']
y_upper = df_yearly_summary['upper_bound']

# Fit models
model_median = LinearRegression().fit(X, y_median)
model_lower = LinearRegression().fit(X, y_lower)
model_upper = LinearRegression().fit(X, y_upper)

# Predict for 2024
year_2024 = np.array([[2024]])
predicted_median = model_median.predict(year_2024)[0]
predicted_lower = model_lower.predict(year_2024)[0]
predicted_upper = model_upper.predict(year_2024)[0]
```

Let me know if you'd like a regional breakdown or visualization.

Envie uma mensagem para o Data Analyst

